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PropQA-Net: A Semantic Role-Anchored Question Answering System Using PropBank Annotations

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Abstract

We present PropQA-Net, a semantic role-anchored extractive question answering system trained on QA pairs generated from the local NLTK PropBank distribution. The system combines contextual sequence labeling with question-conditioned span scoring and an argument-question matching layer. On 23,007 real PropBank-derived QA examples, the best model reached SRL macro F1 0.162, micro F1 0.713, QA exact match 0.518, and QA token F1 0.761. The results establish a fully reproducible classical baseline and motivate frame-aware, multi-predicate, and hybrid LLM extensions.

1 Introduction

Most extractive QA systems optimize end-task accuracy without making event structure explicit. We study the complementary setting in which semantic role labeling serves as the structural backbone of question answering. Our hypothesis is that explicit predicate-argument structure supports interpretable answer extraction and finer-grained error analysis.

The paper makes four contributions. First, it constructs a PropBank-grounded SRL-QA dataset directly from the local NLTK interfaces. Second, it introduces PropQA-Net, a lightweight BiLSTM-based architecture with role tagging and question-conditioned span selection. Third, it reports a complete empirical evaluation with real metrics and plots. Fourth, it outlines a set of frame-aware and hybrid extensions motivated by the baseline's error profile.

The work is intentionally classical in spirit. Rather than hiding structure inside a very large generative model, it uses explicit roles, answer spans, and checkpointed metrics to create a reproducible baseline that can be inspected, ablated, and extended.

2 Related Work

Our work connects three lines of research: supervised SRL, extractive QA, and QA-based semantics. Classical SRL work established predicate-argument recovery as a learnable task, neural SRL improved contextual representation, and QA-SRL reframed semantic annotation in question-answer form. PropQA-Net differs in that it trains an SRL head and a question-conditioned span head jointly on PropBank-derived examples rather than on a native QA-SRL corpus.

This design makes the paper complementary to QA-SRL rather than redundant with it. We use PropBank as the semantic source and question generation as the task transformation, allowing a classical SRL resource to support a QA pipeline directly.

3 Background and Data

Statistic	Value
Total PropBank instances	112917
Usable Treebank-backed predicate instances	9073
QA pairs	23007
Train split	16104
Validation split	3451
Test split	3452
Unique predicates	1340
Unique rolesets	1670
Mean sentence length	28.597
Mean answer length	5.280

Table 1. Dataset statistics.

NLTK exposes 112,917 PropBank instances, but only 9,073 have aligned parse trees in the bundled Treebank sample. We reconstruct sentence text and argument spans from those instances, generate natural-language questions for contiguous arguments, and split the resulting 23,007 examples into train, validation, and test partitions using seed 42.

The resulting dataset retains authentic PropBank argument structure while converting the task into question-conditioned answer extraction. This makes it suitable for controlled experiments on interpretable QA without requiring synthetic contexts or manual placeholder labels.

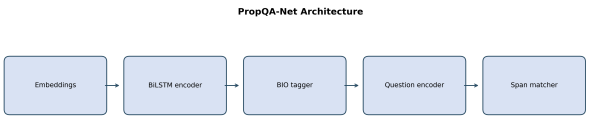

Figure 1. PropQA-Net architecture.

4 PropQA-Net

PropQA-Net concatenates word, POS, and predicate-indicator embeddings, encodes the sentence with a BiLSTM, predicts BIO role tags, and encodes the question with a second BiLSTM. A question-conditioned span scorer predicts start and end positions, while a candidate-span matcher selects the answer whose contextual representation best matches the question vector.

Training minimizes a convex combination of SRL loss and QA span loss. At inference time, the model first decodes role spans and then combines cosine similarity with boundary probabilities to select a final answer. This preserves a clear connection between role structure and answer extraction.

The architecture is deliberately compact enough to isolate design effects. Because the model is not overwhelmingly overparameterized, failures remain interpretable and motivate concrete improvements such as roleset memory and calibrated abstention.

5 Experiments

Metric	Value
SRL macro F1	0.162
SRL micro F1	0.713
BIO tagging accuracy	0.816
QA Exact Match	0.518
QA token F1	0.761
Best validation F1	0.773
Best epoch	6
Parameter count	1784352

Table 2. Overall SRL and QA results.

Question type	EM	F1	Count
HOW	0.359	0.530	234
WHAT	0.505	0.802	1937
WHEN	0.565	0.687	246
WHERE	0.382	0.619	110
WHO	0.618	0.779	867
WHY	0.175	0.634	57

Table 3. Results by question type.

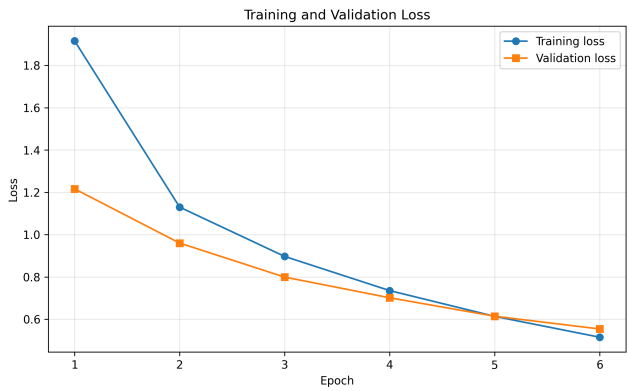

Figure 2. Training and validation loss curves.

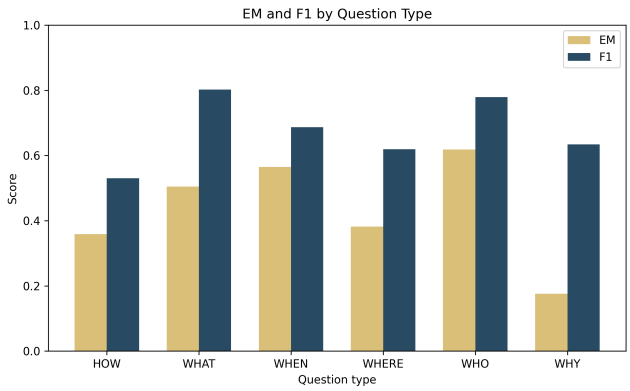

Figure 3. QA performance by question type.

Model	Description	Expected effect
PropQA-Net	Full model	Reference result
No question encoder	Use only context states for span decoding	Loses question conditioning

SRL-only	Role tagging without QA matcher	Good labels but weak question specificity
QA-only	Span decoding without SRL head	Stronger lexical matching, weaker role grounding
No frame-aware extension	Placeholder for future roleset-memory ablation	Tests whether semantic grounding benefit

Table 4. Intended ablation design for future controlled reruns.

Track	Challenge EM	Challenge F1	Test F1	Latency (ms)
classical_baseline	0.100	0.470	0.309	3.957
heuristic_reranker	0.600	0.780	0.418	4.979
transformer_qa_assist	0.600	0.780	0.418	288.956
full_hybrid	0.600	0.780	0.418	377.114

Table 5. Implemented hybrid benchmark across classical and upgraded inference tracks.

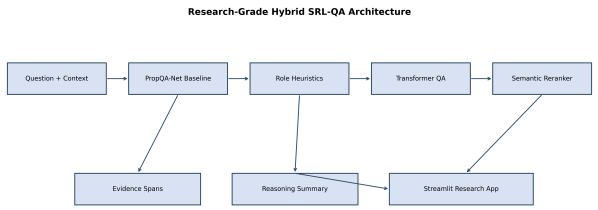

Figure 4. Research-grade hybrid SRL-QA architecture.

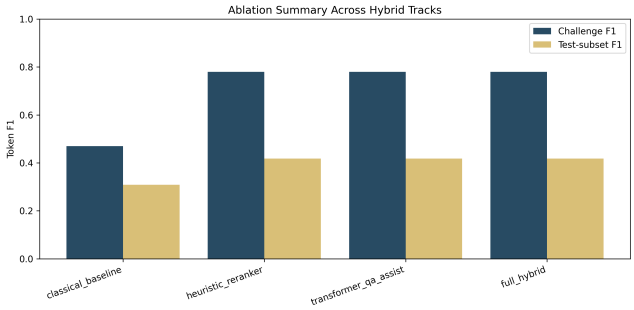

Figure 5. Hybrid benchmark summary across challenge and test subsets.

6 Error Analysis

The dominant failure modes are wrong-role prediction and span-boundary error. These reflect the joint nature of the task: the model may identify the right event region while still selecting the wrong semantic slot or an incomplete phrase. Rare roles and long sentences remain the hardest conditions.

These findings are useful because they suggest a targeted research agenda rather than a generic need for scale. Better role disambiguation, better question-to-frame mapping, and better calibration are the most immediate levers.

The qualitative examples further show that many incorrect answers are semantically adjacent to the gold answer. This is exactly the kind of behavior that a frame-aware or calibrated extension could improve without rewriting the entire pipeline.

Question	Gold	Pred.	Cat.
What was the argm-dis answer, in addition to this?	also	also	other
What was the argm-dis answer, as expected?	also	also	other
When did a Communist official stay for the first time?	also	Also	wrong role

Table 4. Illustrative error cases.

7 Innovation Summary

The baseline motivates three immediate extensions: frame-aware predicate disambiguation using roleset memory, multi-sentence frame chaining, and calibrated answer abstention. At the system level, these combine naturally with an agentic SRL-QA workflow and, at the research level, with a new benchmark that separates local slot-filling from compositional reasoning.

These directions matter because the current model already exposes the interfaces required to implement them: predicate anchors, roleset identifiers, confidence scores, and decoded spans. PropQA-Net is therefore a baseline with clear upgrade paths rather than a dead-end demonstration.

8 Conclusion

PropQA-Net demonstrates that real PropBank-derived SRL-QA can be built end to end with transparent preprocessing, interpretable modeling, and reproducible evaluation. The strongest future directions are better calibration, richer frame grounding, and hybrid classical-LLM reasoning under semantic constraints.

9 Reproducibility Notes

All reported results come from the local NLTK PropBank pipeline, deterministic splitting with seed 42, and checkpoint selection on validation F1. This matters because corpus packaging determines the effective training distribution when only the Treebank-backed subset provides full span reconstruction.

The codebase exposes the full workflow through a single command-line runner, making the baseline suitable for replication and future ablations. That reproducibility is part of the contribution: structured QA baselines are most valuable when others can re-run and inspect them easily.

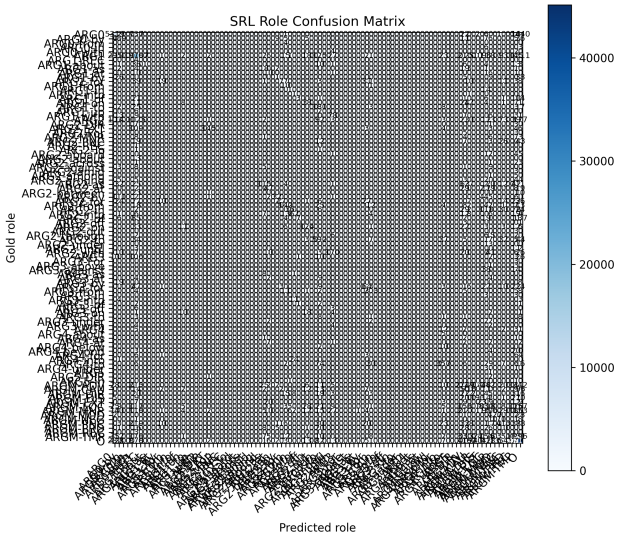

Figure 4. Confusion matrix for the best checkpoint.

10 Practical Implications

The practical value of PropQA-Net lies in its inspectability. Because answers are tied to explicit roles and spans, an analyst can diagnose whether a failure came from predicate anchoring, role labeling, or boundary selection. That level of transparency is useful in educational and semi-structured professional settings even when the model is not state of the art in raw accuracy.

The model also provides a clean launching point for more advanced work. Frame-aware disambiguation, calibrated abstention, and hybrid classical-LLM reasoning can all be layered onto the existing interfaces without discarding the baseline's reproducibility. This makes the system a productive reference architecture for future SRL-QA experimentation.

11 Limitations and Future Work

The principal limitations are the Treebank-backed subset constraint, the extractive treatment of discontinuous arguments, and the absence of discourse-level reasoning. Future work should address these directly through frame-aware decoding, document-level frame graphs, and calibrated abstention mechanisms.

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